

NORTH WEST SHELF ANALYSIS

FEISTY spatial yield analysis for the North West Shelf

Spawning-stock biomass, yield change and historical catch comparison

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24 August 2026

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1. Purpose

This document presents the spatial FEISTY analysis of spawning-stock biomass (SSB) and yield in four North West Shelf study regions: Bay of Biscay, Celtic Sea, English Channel and North Sea.

Mean conditions in 2090-2100 are compared with the 2020-2029 reference period. Results are shown for the total of the three FEISTY fishing groups and separately for small pelagics, demersals and large pelagics. Every selected fishing strategy must keep the SSB of each group at or above 30% of its unfished value (SSB0).

2. Reproducibility and project structure

The complete analysis is documented in the Quarto file `NWS_FEISTY_spatial_yield_analysis.qmd`. All paths are relative to the repository root. The two large FEISTY inputs are not stored on GitHub because of their size; they are required only when the full spatial simulation is rerun.

- `data/nws_for_feisty_proj_year.RDS`
- `data/all_FFMSY.RDS`

The repository already contains the cell classification, selected strategies, period-level model output, validation tables, historical catch comparison and final figures. The report can therefore be rendered without repeating the 22,317-cell run.

```
source(file.path(base_dir, "NWS_FINAL_maps_FULL_22317_cells.R"), chdir = TRUE)
```

3. Study regions and spatial grid

The final calculation uses every available environmental cell within the four study regions. The same approximate longitude-latitude boxes are retained from the earlier FEISTY workflow.

Region	Number of cells
Bay of Biscay	5,873
Celtic Sea	1,904
English Channel	1,346
North Sea	13,194

Table 1. Spatial cells included in the final analysis.

4. Chosen fishing mortalities and the 30% SSB0 constraint

For each region and period, the maximum-yield strategy that passed the initial regional 30% SSB0 criterion was selected. The constraint was then recalculated after the spatial simulations using spatial mean SSB and SSB0 values.

Bay of Biscay required a small reduction in fishing intensity after the initial spatial validation. The final values are a constant F of 0.33 for 2020-2029 and a temperature-scaled FFMSY coefficient of 0.54 for 2090-2100.

Region	Period	Approach	F / coefficient	Minimum SSB / SSB0	Limiting group
Bay of Biscay	2020-2029	Constant F	0.33	0.303	Large pelagics
Bay of Biscay	2090-2100	Temp.-scaled FFMSY	0.54	0.307	Demersals
Celtic Sea	2020-2029	Constant F	0.12	0.309	Large pelagics
Celtic Sea	2090-2100	Constant F	0.12	0.322	Large pelagics
English Channel	2020-2029	Constant F	0.15	0.302	Large pelagics
English Channel	2090-2100	Constant F	0.15	0.306	Large pelagics
North Sea	2020-2029	Temp.-scaled FFMSY	0.37	0.317	Large pelagics
North Sea	2090-2100	Constant F	0.17	0.319	Large pelagics

Table 2. Selected regional fishing strategy and fishing value.

5. Final full-grid 30% SSB0 validation

The final results are accepted only when every selected regional strategy passes the full-grid validation. All eight region-period combinations satisfy the management rule.

Region	Period	Small pel.	Demersals	Large pel.	Minimum	Limiting group	Valid
Bay of Biscay	2020-2029	0.960	0.648	0.308	0.308	Large pelagics	Yes
Bay of Biscay	2090-2100	1.011	0.434	0.304	0.304	Large pelagics	Yes
Celtic Sea	2020-2029	1.029	0.831	0.454	0.454	Large pelagics	Yes
Celtic Sea	2090-2100	1.031	0.877	0.468	0.468	Large pelagics	Yes
English Channel	2020-2029	1.029	0.823	0.557	0.557	Large pelagics	Yes
English Channel	2090-2100	1.041	0.840	0.570	0.570	Large pelagics	Yes
North Sea	2020-2029	0.887	0.623	0.319	0.319	Large pelagics	Yes
North Sea	2090-2100	0.975	0.721	0.382	0.382	Large pelagics	Yes

Table 3. Full-grid SSB / SSB0 validation of the selected strategies.

6. Spatial SSB and yield results

The beginning of the analysis period is represented by the 2020-2029 mean. Absolute change is calculated as the 2090-2100 mean minus the 2020-2029 mean. Total SSB and total yield are the sums of small pelagics, demersals and large pelagics.

6.1. SSB at the beginning of the period

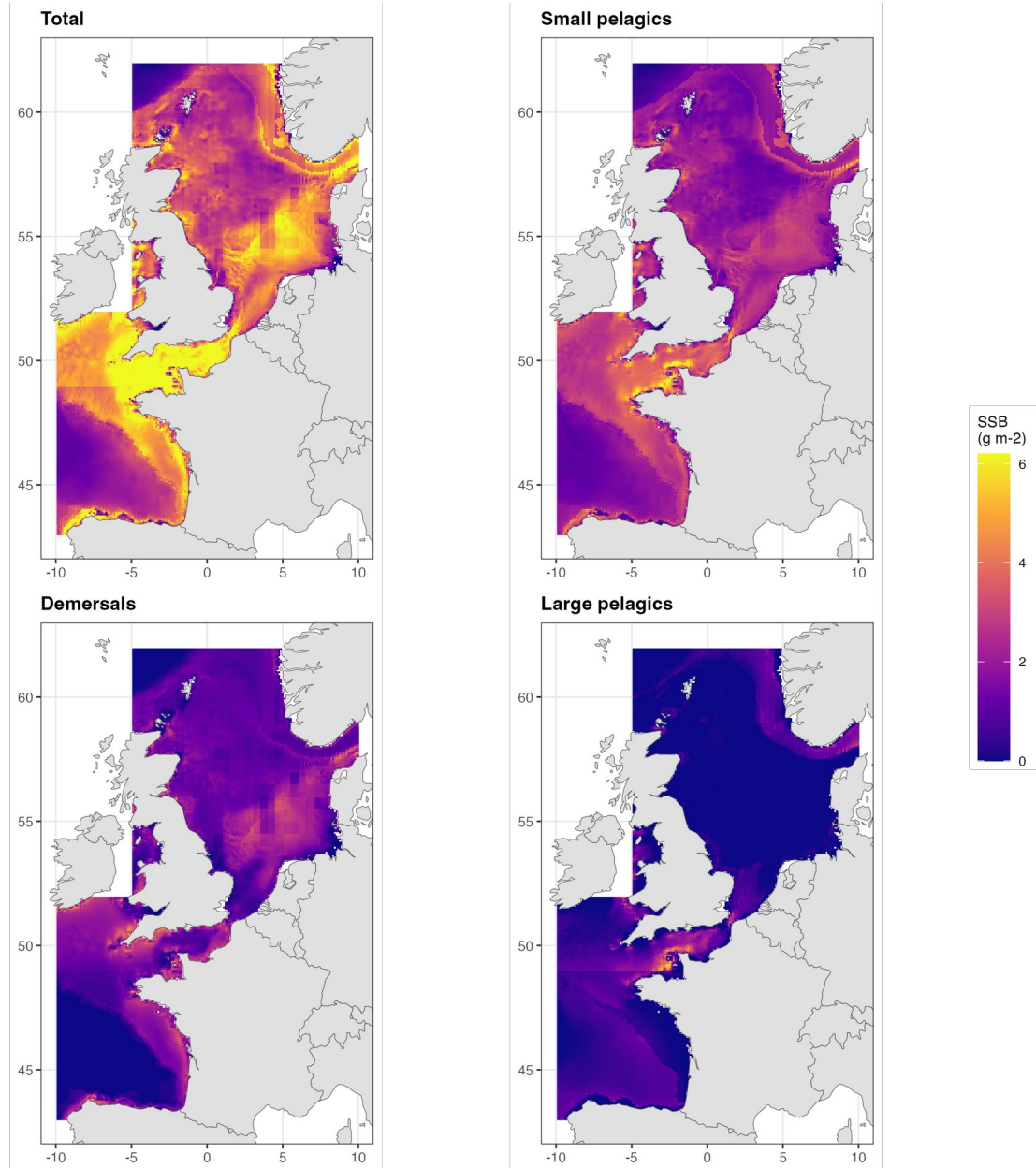


Figure 1. Mean spawning-stock biomass in 2020-2029 under the selected regional fishing strategies.

Values are in g m⁻². All four panels use the same colour scale, capped at the pooled 98th percentile.

6.2. Yield at the beginning of the period

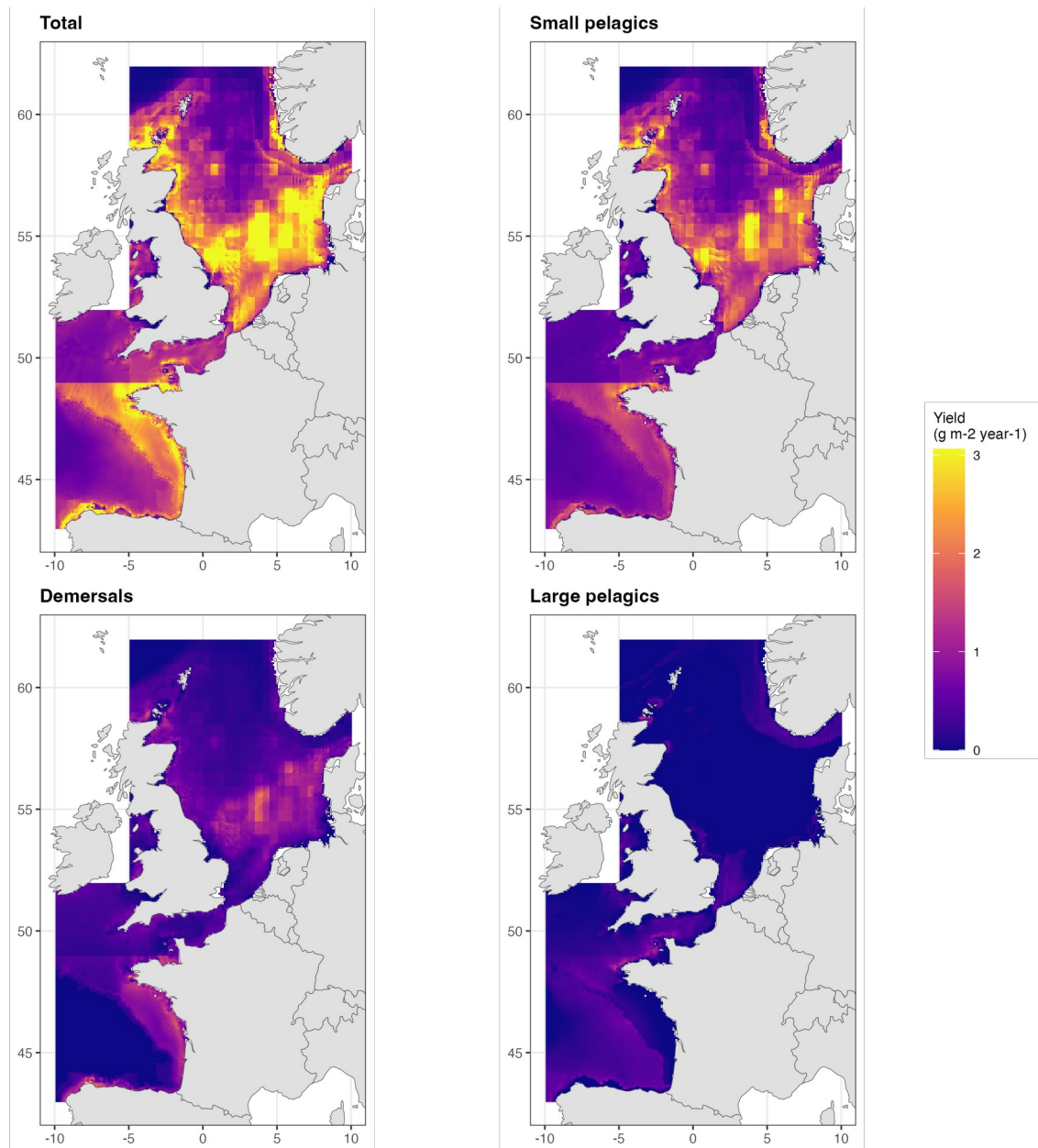


Figure 2. Mean yield in 2020-2029 under the selected regional fishing strategies.

Values are in g m⁻² year⁻¹. All four panels use the same colour scale, capped at the pooled 98th percentile.

6.3. Absolute change in SSB

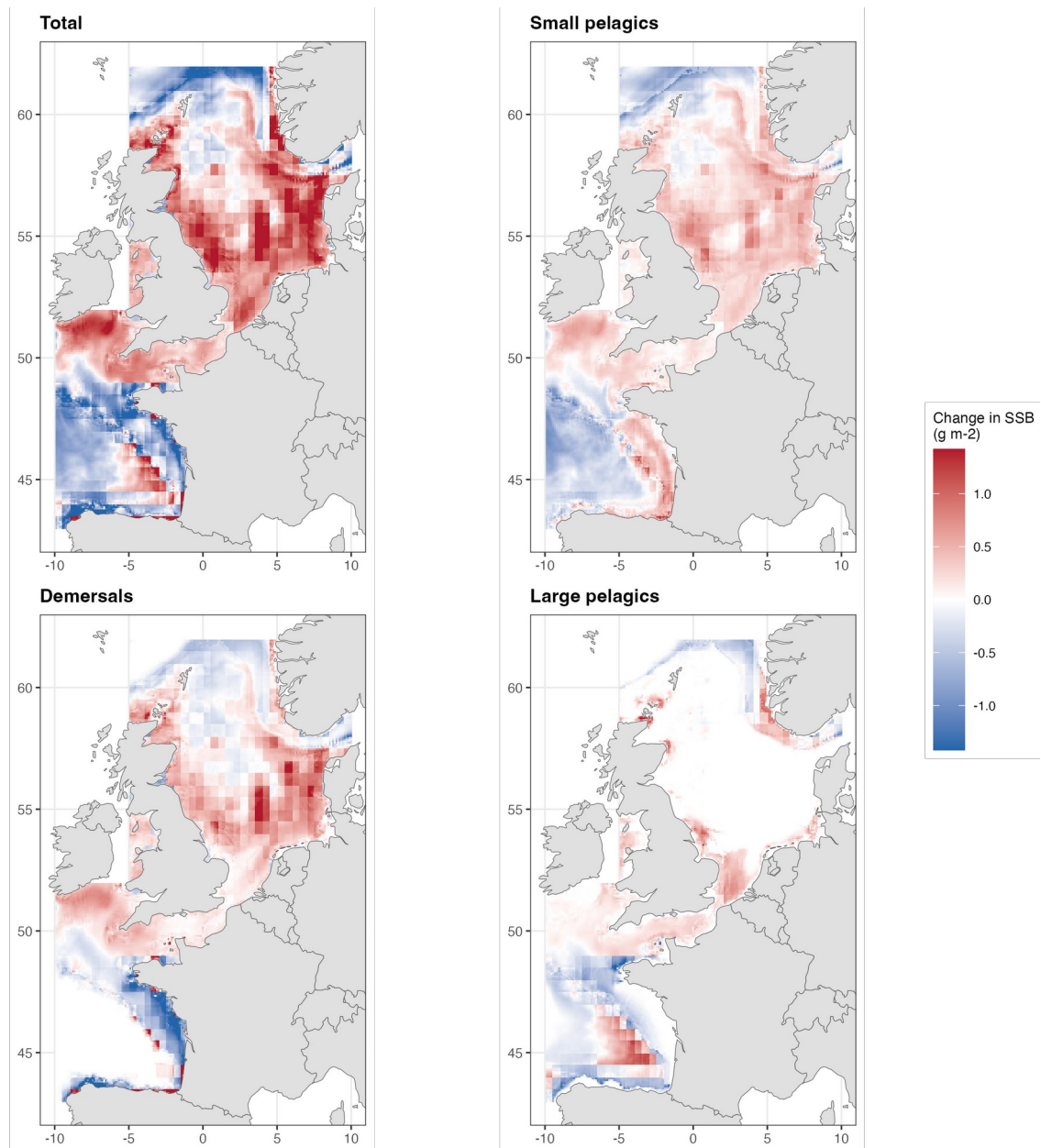


Figure 3. Absolute SSB change: mean 2090-2100 minus mean 2020-2029.

Values are in g m^{-2} . All panels share the same symmetric colour scale; blue indicates a decrease and red an increase.

6.4. Absolute change in yield

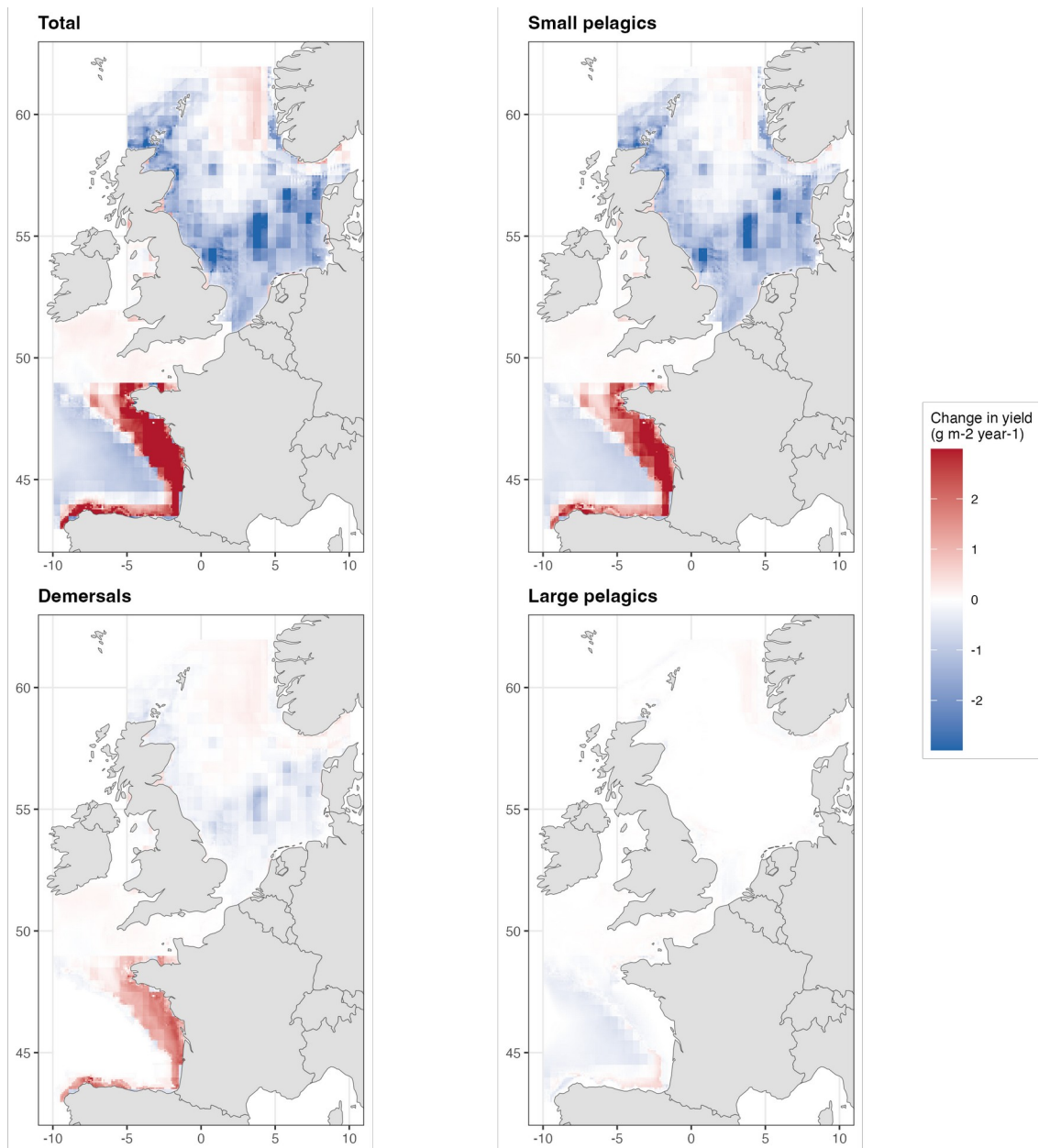
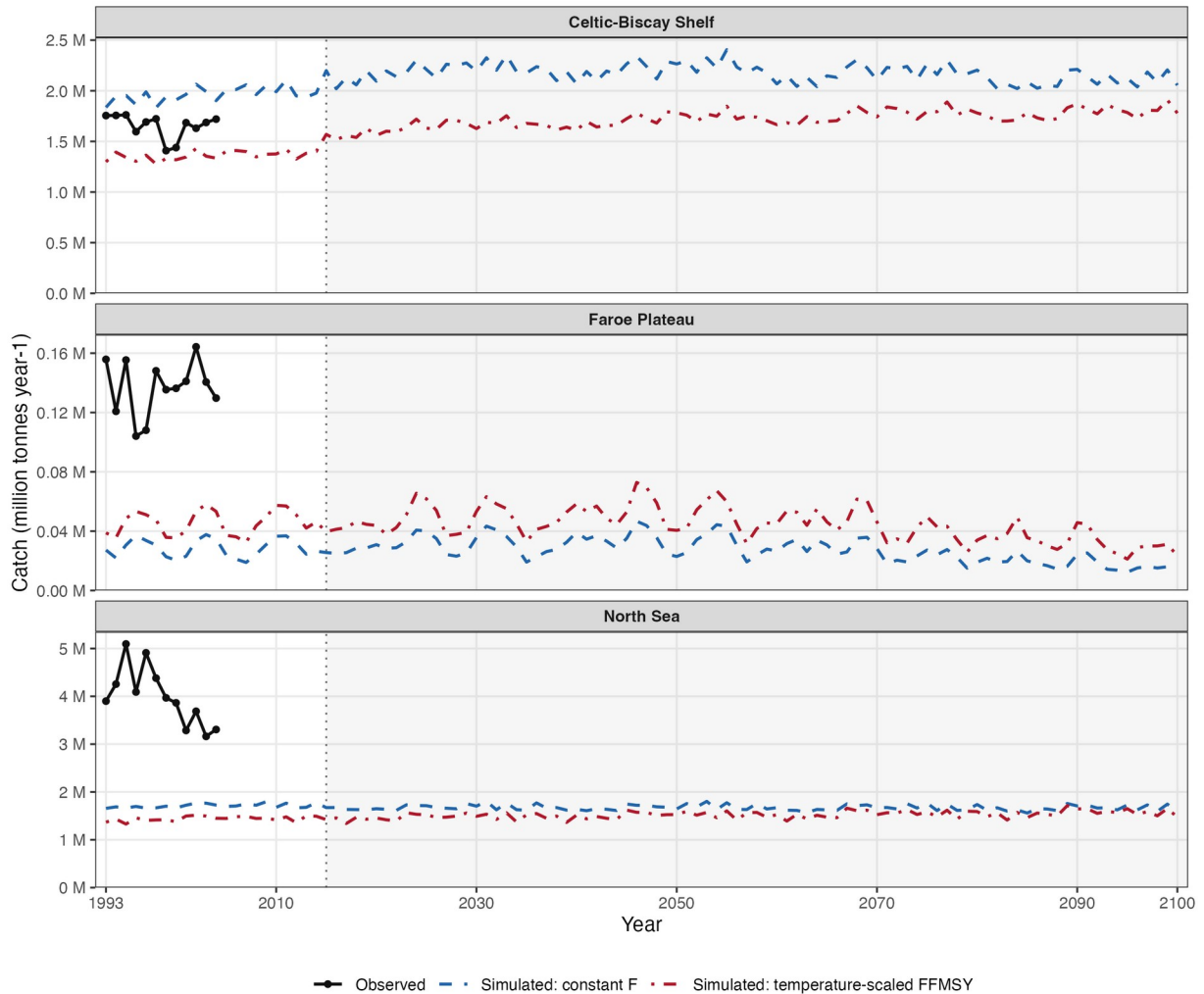


Figure 4. Absolute yield change: mean 2090-2100 minus mean 2020-2029.

Values are in g m⁻² year⁻¹. All panels share the same symmetric colour scale; blue indicates a decrease and red an increase.

7. Observed and simulated catch

Observed catch is available for 1993-2004. The constant-F and temperature-scaled FFMSY simulations extend to 2100, using historical forcing through 2014 and SSP3-7.0 projection forcing from 2015.



Dotted vertical line: start of the SSP3-7.0 projection forcing.

Figure 5. Observed total catch (1993-2004) and simulations (1993-2100) in the North Sea, Celtic-Biscay Shelf and Faroe Plateau.

Each ecosystem has its own vertical scale, beginning at zero; the dotted line marks the start of projection forcing in 2015.

8. Regional yield-change summary

The table summarises mean cell-level yield changes by region and group. These averages should be interpreted together with the maps because changes are spatially heterogeneous.

Region	Group	Mean absolute change	Mean change (%)	Cells decreasing (%)
Bay of Biscay	Total yield	0.495	-4.29	60.6
Bay of Biscay	Small pelagics	0.308	1.50	59.9
Bay of Biscay	Demersals	0.304	455.82	60.7
Bay of Biscay	Large pelagics	-0.116	-64.77	89.7
Celtic Sea	Total yield	0.084	9.74	16.2
Celtic Sea	Small pelagics	0.034	7.41	16.6
Celtic Sea	Demersals	0.040	11.64	20.5
Celtic Sea	Large pelagics	0.010	5.24	25.8
English Channel	Total yield	0.071	5.85	2.0
English Channel	Small pelagics	0.024	3.72	6.6
English Channel	Demersals	0.025	27.88	9.9
English Channel	Large pelagics	0.022	-4.72	26.6
North Sea	Total yield	-0.715	-9.22	79.9
North Sea	Small pelagics	-0.632	-3.94	86.7
North Sea	Demersals	-0.085	161.80	67.3
North Sea	Large pelagics	0.002	810.61	55.7

Table 4. Regional summary of cell-level yield change.

9. Interpretation and limitations

- The maps represent SSB and yield under regionally selected fishing strategies; fishing mortality is not optimised separately in each cell.
- The beginning-period maps show the mean for 2020-2029 rather than a single initial year.
- Total values are the sums of small pelagics, demersals and large pelagics.
- The 30% threshold is checked for each group using full-grid spatial means.
- Observed catch is available for 1993-2004; the corresponding simulations continue to 2100 for the three validation ecosystems.
- The four projection regions are defined by approximate longitude-latitude boxes retained from the original workflow.

10. Software

The analysis was run in R using the FEISTY, tidyverse, data.table, maps and cowplot packages. Rendering requires Quarto and knitr. The full script records the model setup, parallel processing, checkpointing, FFMSY matching, SSB and yield calculations, the SSB0 validation, tables and figures.